

Digital Electronics Question and Answers

Flip Flops-2

Topic 7

Flip Flops-2

- 1.) D Flip Flop
- 2.) Triggering of Flip Flops
- 3.) Master-Slave Flip-Flops
- 4.) Realisation of one Flip-flop using other Flip-flops

Digital Electronics Question & Answers

1.) D Flip Flop

1. In D flip-flop, D stands for _____

- a) Distant
- b) Data
- c) Desired
- d) Delay

Answer: b

Explanation: The D of D-flip-flop stands for “data”. It stores the value on the data line.

2. The D flip-flop has _____ input.

- a) 1
- b) 2
- c) 3
- d) 4

Answer: a

Explanation: The D flip-flop has one input. The D of D-flip-flop stands for “data”. It stores the value on the data line.

3. The D flip-flop has _____ output/outputs.

- a) 2
- b) 3
- c) 4
- d) 1

Answer: a

Explanation: The D flip-flop has two outputs: Q and Q complement. The D flip-flop has one input. The D of D-flip-flop stands for “data”. It stores the value on the data line.

4. A D flip-flop can be constructed from an _____ flip-flop.

- a) S-R
- b) J-K
- c) T
- d) S-K

Answer: a

Explanation: A D flip-flop can be constructed from an S-R flip-flop by inserting an inverter between S and R and assigning the symbol D to the S input.

5. In D flip-flop, if clock input is LOW, the D input _____

- a) Has no effect
- b) Goes high
- c) Goes low
- d) Has effect

Answer: a

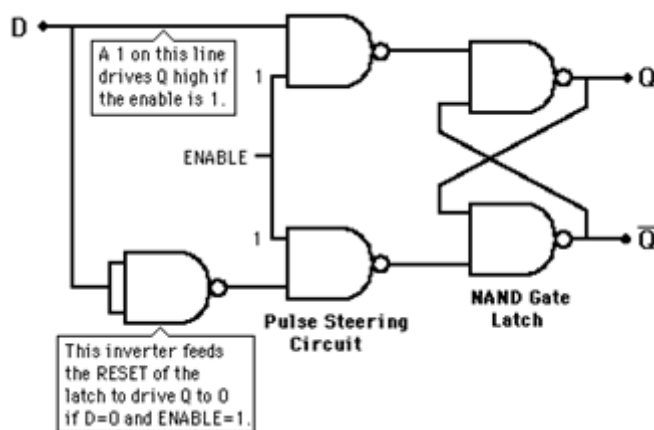
Explanation: In D flip-flop, if clock input is LOW, the D input has no effect, since the set and reset inputs of the NAND flip-flop are kept HIGH.

6. In D flip-flop, if clock input is HIGH & D=1, then output is _____

- a) 0
- b) 1
- c) Forbidden
- d) Toggle

Answer: a

Explanation: If clock input is HIGH & D=1, then output is 0. It can be observed from this diagram:



7. Which statement describes the BEST operation of a negative-edge-triggered D flip-flop?

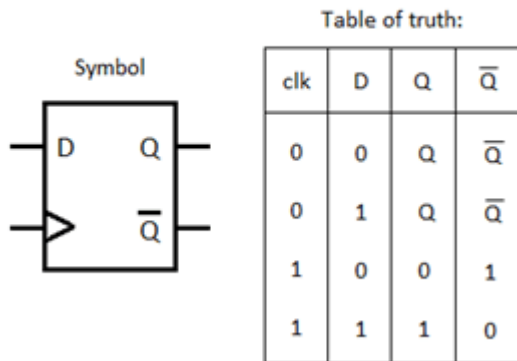
- a) The logic level at the D input is transferred to Q on NGT of CLK
- b) The Q output is ALWAYS identical to the CLK input if the D input is HIGH
- c) The Q output is ALWAYS identical to the D input when CLK = PGT
- d) The Q output is ALWAYS identical to the D input

Answer: a

Explanation: By the truth table of D flip flop, we can observe that Q always depends on D.

Hence, for every negative trigger pulse, the logic at input D is shifted to Output Q.

D Flip-flop



8. Which of the following is correct for a gated D flip-flop?

- a) The output toggles if one of the inputs is held HIGH
- b) Only one of the inputs can be HIGH at a time
- c) The output complement follows the input when enabled
- d) Q output follows the input D when the enable is HIGH

Answer: d

Explanation: If clock is high then the D flip-flop operate and we know that input is equals to output in case of D flip-flop. It stores the value on the data line.

9. With regard to a D latch _____

- a) The Q output follows the D input when EN is LOW
- b) The Q output is opposite the D input when EN is LOW
- c) The Q output follows the D input when EN is HIGH
- d) The Q output is HIGH regardless of EN's input state

Answer: c

Explanation: Latch is nothing but flip flop which holds the o/p or i/p state. And in D flip-flop output follows the input. It stores the value on the data line.

10. Which of the following is correct for a D latch?

- a) The output toggles if one of the inputs is held HIGH
- b) Q output follows the input D when the enable is HIGH
- c) Only one of the inputs can be HIGH at a time
- d) The output complement follows the input when enabled

Answer: b

Explanation: If the clock is HIGH then the D flip-flop operates and we know that input equals to output in case of D flip flop. It stores the value on the data line.

11. Which of the following describes the operation of a positive edge-triggered D flip-flop?

- a) If both inputs are HIGH, the output will toggle
- b) The output will follow the input on the leading edge of the clock
- c) When both inputs are LOW, an invalid state exists
- d) The input is toggled into the flip-flop on the leading edge of the clock and is passed to the output on the trailing edge of the clock

Answer: b

Explanation: Edge-triggered flip-flop means the device will change state during the rising or falling edge of the clock pulse. The main phenomenon of the D flip-flop is that the o/p will follow the i/p when the enable pin is HIGH.

12. A D flip-flop utilizing a PGT clock is in the CLEAR state. Which of the following input actions will cause it to change states?

- a) CLK = NGT, D = 0
- b) CLK = PGT, D = 0
- c) CLOCK NGT, D = 1
- d) CLOCK PGT, D = 1

Answer: d

Explanation: PGT refers to Positive Going Transition and NGT refers to negative Going Transition. Earlier, the DFF is in a clear state (output is 0). So, if D = 1 then in the next stage output will be 1 and hence the stage will be changed.

13. A positive edge-triggered D flip-flop will store a 1 when _____

- a) The D input is HIGH and the clock transitions from HIGH to LOW
- b) The D input is HIGH and the clock transitions from LOW to HIGH
- c) The D input is HIGH and the clock is LOW
- d) The D input is HIGH and the clock is HIGH

Answer: b

Explanation: A positive edge-triggered D flip-flop will store a 1 when the D input is HIGH and the clock transitions from LOW to HIGH. While a negative edge-triggered D flip-flop will store a 0 when the D input is HIGH and the clock transitions from HIGH to LOW.

14. Why do the D flip-flops receive its designation or nomenclature as 'Data Flip-flops'?

- a) Due to its capability to receive data from flip-flop
- b) Due to its capability to store data in flip-flop
- c) Due to its capability to transfer the data into flip-flop
- d) Due to erasing the data from the flip-flop

Answer: c

Explanation: Due to its capability to transfer the data into flip-flop. D-flip-flops stores the value on the data line.

15. The characteristic equation of D-flip-flop implies that _____

- a) The next state is dependent on previous state
- b) The next state is dependent on present state
- c) The next state is independent of previous state
- d) The next state is independent of present state

Answer: d

Explanation: A characteristic equation is needed when a specific gate requires a specific output in order to satisfy the truth table. The characteristic equation of D flip-flop is given by $Q(n+1) = D$; which indicates that the next state is independent of the present state.

2.) Triggering of Flip Flops

1. The characteristic equation of J-K flip-flop is _____

- a) $Q(n+1) = JQ(n) + K'Q(n)$
- b) $Q(n+1) = J'Q(n) + KQ'(n)$
- c) $Q(n+1) = JQ'(n) + KQ(n)$
- d) $Q(n+1) = JQ'(n) + K'Q(n)$

Answer: d

Explanation: A characteristic equation is needed when a specific gate requires a specific output in order to satisfy the truth table. The characteristic equation of J-K flip-flop is given by: $Q(n+1) = JQ'(n) + K'Q(n)$.

2. In a J-K flip-flop, if $J=K$ the resulting flip-flop is referred to as _____

- a) D flip-flop
- b) S-R flip-flop
- c) T flip-flop
- d) S-K flip-flop

Answer: c

Explanation: In J-K flip-flop, if both the inputs are same then it behaves like T flip-flop.

3. In J-K flip-flop, the function $K=J$ is used to realize _____

- a) D flip-flop
- b) S-R flip-flop
- c) T flip-flop
- d) S-K flip-flop

Answer: c

Explanation: T flip-flop allows the same inputs. So, in J-K flip-flop $J=K$ then it will work as T flip-flop.

4. The only difference between a combinational circuit and a flip-flop is that _____

- a) The flip-flop requires previous state
- b) The flip-flop requires next state
- c) The flip-flop requires a clock pulse
- d) The flip-flop depends on the past as well as present states

Answer: d

Explanation: Both flip-flop and latches are memory elements with clock/control inputs. They depend on the past as well as present states. Whereas, in case of combinational circuits, they only depend on the present state.

5. How many stable states combinational circuits have?

- a) 3
- b) 4
- c) 2
- d) 5

Answer: c

Explanation: The two stable states of combinational circuits are 1 and 0. Whereas, in flip-flops there is an additional state known as Forbidden State.

6. The flip-flop is only activated by _____

- a) Positive edge trigger
- b) Negative edge trigger
- c) Either positive or Negative edge trigger
- d) Sinusoidal trigger

Answer: c

Explanation: Flip flops can be activated with either a positive or negative edge trigger.

7. The S-R latch composed of NAND gates is called an active low circuit because _____

- a) It is only activated by a positive level trigger
- b) It is only activated by a negative level trigger
- c) It is only activated by either a positive or negative level trigger
- d) It is only activated by sinusoidal trigger

Answer: b

Explanation: Active low indicates that only an input value of 0 sets or resets the circuit.

8. Both the J-K & the T flip-flop are derived from the basic _____

- a) S-R flip-flop
- b) S-R latch

- c) D latch
- d) D flip-flop

Answer: b

Explanation: The SR latch is the basic block for the D latch/flip flop from which the JK and T flip flops are derived. A latch is similar to a flip-flop, only without a clock input.

9. The flip-flops which has not any invalid states are _____

- a) S-R, J-K, D
- b) S-R, J-K, T
- c) J-K, D, S-R
- d) J-K, D, T

Answer: d

Explanation: Unlike the SR latch, these circuits have no invalid states. The SR latch or flip-flop has an invalid or forbidden state where no output could be determined.

10. What does the triangle on the clock input of a J-K flip-flop mean?

- a) Level enabled
- b) Edge triggered
- c) Both Level enabled & Edge triggered
- d) Level triggered

Answer: b

Explanation: The triangle on the clock input of a J-K flip-flop mean edge triggered. Whereas the absence of triangle symbol implies that the flip-flop is level-triggered.

11. What does the circle on the clock input of a J-K flip-flop mean?

- a) Level enabled
- b) Positive edge triggered
- c) negative edge triggered
- d) Level triggered

Answer: c

Explanation: The circle on the clock input of a J-K flip-flop mean negative edge triggered. Whereas the absence of triangle symbol implies that the flip-flop is level-triggered.

12. What does the direct line on the clock input of a J-K flip-flop mean?

- a) Level enabled
- b) Positive edge triggered
- c) negative edge triggered
- d) Level triggered

Answer: d

Explanation: The direct line on the clock input of a J-K flip-flop mean level triggered. Whereas the presence of triangle symbol implies that the flip-flop is edge-triggered.

13. What does the half circle on the clock input of a J-K flip-flop mean?

- a) Level enabled
- b) Positive edge triggered
- c) negative edge triggered
- d) Level triggered

Answer: d

Explanation: The half circle on the clock input of a J-K flip-flop mean level triggered. Whereas the presence of triangle symbol implies that the flip-flop is edge-triggered.

14. A J-K flip-flop with $J = 1$ and $K = 1$ has a 20 kHz clock input. The Q output is _____

- a) Constantly LOW
- b) Constantly HIGH
- c) A 20 kHz square wave
- d) A 10 kHz square wave

Answer: d

Explanation: As one flip flop is used so there are two states available. So, $20/2 = 10\text{Hz}$ frequency is available at the output.

15. On a positive edge-triggered S-R flip-flop, the outputs reflect the input condition when _____

- a) The clock pulse is LOW
- b) The clock pulse is HIGH
- c) The clock pulse transitions from LOW to HIGH
- d) The clock pulse transitions from HIGH to LOW

Answer: c

Explanation: Edge triggered device will follow the input condition when there is a transition. It is said to be positive edge triggered when transition occurs from LOW to HIGH. While it is said to be a negative edge triggered when a transition occurs from HIGH to LOW.

3.) Master-Slave Flip-Flops

1. The asynchronous input can be used to set the flip-flop to the _____

- a) 1 state
- b) 0 state

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- c) either 1 or 0 state
- d) forbidden State

Answer: c

Explanation: The asynchronous input can be used to set the flip-flop to the 1 state or clear the flip-flop to the 0 state at any time, regardless of the condition at the other inputs.

2. Input clock of RS flip-flop is given to _____

- a) Input
- b) Pulser
- c) Output
- d) Master slave flip-flop

Answer: b

Explanation: Pulser behaves like an arithmetic operator, to perform the operation or determination of corresponding states.

3. D flip-flop is a circuit having _____

- a) 2 NAND gates
- b) 3 NAND gates
- c) 4 NAND gates
- d) 5 NAND gates

Answer: c

Explanation: D flip-flop is a circuit having 4 NAND gates. Two of them are connected with each other.

4. In JK flip flop same input, i.e. at a particular time or during a clock pulse, the output will oscillate back and forth between 0 and 1. At the end of the clock pulse the value of output Q is uncertain. The situation is referred to as?

- a) Conversion condition
- b) Race around condition
- c) Lock out state
- d) Forbidden State

Answer: b

Explanation: A race around condition is a flaw in an electronic system or process whereby the output and result of the process is unexpectedly dependent on the sequence or timing of other events.

5. Master slave flip flop is also referred to as?

- a) Level triggered flip flop
- b) Pulse triggered flip flop
- c) Edge triggered flip flop

d) Edge-Level triggered flip flop

Answer: b

Explanation: The term pulse triggered means the data is entered on the rising edge of the clock pulse, but the output does not reflect the change until the falling edge of the clock pulse.

6. In a positive edge triggered JK flip flop, a low J and low K produces?

- a) High state
- b) Low state
- c) Toggle state
- d) No Change State

Answer: d

Explanation: In JK Flip Flop if $J = K = 0$ then it holds its current state. There will be no change.

7. If one wants to design a binary counter, the preferred type of flip-flop is _____

- a) D type
- b) S-R type
- c) Latch
- d) J-K type

Answer: d

Explanation: If one wants to design a binary counter, the preferred type of flip-flop is J-K type because it has capability to recover from toggle condition. SR flip-flop is not suitable as it produces the "Invalid State".

8. S-R type flip-flop can be converted into D type flip-flop if S is connected to R through _____

- a) OR Gate
- b) AND Gate
- c) Inverter
- d) Full Adder

Answer: c

Explanation: S-R type flip-flop can be converted into D type flip-flop if S is connected to R through an Inverter gate.

9. Which of the following flip-flops is free from the race around the problem?

- a) T flip-flop
- b) SR flip-flop
- c) Master-Slave Flip-flop
- d) D flip-flop

Answer: a

Explanation: T flip-flop is free from the race around condition because its output depends only on the input; hence there is no any problem creates as like toggle.

10. Which of the following is the Universal Flip-flop?

- a) S-R flip-flop
- b) J-K flip-flop
- c) Master slave flip-flop
- d) D Flip-flop

Answer: b

Explanation: There are lots of flip-flops can be prepared by using J-K flip-flop. So, the name is a universal flip-flop. Also, the JK flip-flop resolves the Forbidden State.

11. How many types of triggering take place in a flip flops?

- a) 3
- b) 2
- c) 4
- d) 5

Answer: a

Explanation: There are three types of triggering in a flip-flop, viz., level triggering, edge triggering and pulse triggering.

12. Flip-flops are _____

- a) Stable devices
- b) Astable devices
- c) Bistable devices
- d) Monostable devices

Answer: c

Explanation: Flip-flops are synchronous bistable devices known as bistable multivibrators as they have 2 stable states.

13. The term synchronous means _____

- a) The output changes state only when any of the input is triggered
- b) The output changes state only when the clock input is triggered
- c) The output changes state only when the input is reversed
- d) The output changes state only when the input follows it

Answer: b

Explanation: The term synchronous means the output changes state only when the clock input is triggered. That is, changes in the output occur in synchronization with the clock.

14. The S-R, J-K and D inputs are called _____

- a) Asynchronous inputs
- b) Synchronous inputs
- c) Bidirectional inputs
- d) Unidirectional inputs

Answer: b

Explanation: The S-R, J-K and D inputs are called synchronous inputs because data on these inputs are transferred to the flip-flop's output only on the triggering edge or level triggering of the clock pulse. Moreover, flip-flops have a clock input whereas latches don't. Hence, known as synchronous inputs.

15. The circuit that generates a spike in response to a momentary change of input signal is called _____

- a) R-C differentiator circuit
- b) L-R differentiator circuit
- c) R-C integrator circuit
- d) L-R integrator circuit

Answer: a

Explanation: The circuit that generates a spike in response to a momentary change of input signal is called R-C differentiator circuit.

4.) Realisation of one Flip-flop using other Flip-flops

1. To realise one flip-flop using another flip-flop along with a combinational circuit, known as _____

- a) PREVIOUS state decoder
- b) NEXT state decoder
- c) MIDDLE state decoder
- d) PRESENT state decoder

Answer: b

Explanation: To realise one flip-flop using another flip-flop along with a combinational circuit, known as NEXT state decoder which acts as like a flip-flop.

2. For realisation of JK flip-flop from SR flip-flop, the input J and K will be given as _____

- a) External inputs to S and R
- b) Internal inputs to S and R
- c) External inputs to combinational circuit
- d) Internal inputs to combinational circuit

Answer: a

Explanation: If a JK Flip Flop is required, the inputs are given to the combinational circuit and the output of the combinational circuit is connected to the inputs of the actual flip flop. So, J and K will be given as external inputs to S and R. As SR flip-flop have invalid state and JK flip-flop don't.

3. For realisation of JK flip-flop from SR flip-flop, if $J=0$ & $K=0$ then the input is _____

- a) $S=0, R=0$
- b) $S=0, R=X$
- c) $S=X, R=0$
- d) $S=X, R=X$

Answer: b

Explanation: If $J=0$ & $K=0$, the output will be as: $Q(n)=0, Q(n+1)=0$ and it is fed into both the AND gates which results as $S=0$ & $R=X$ (i.e. don't care).

FF data inputs		Output	S-R FF inputs	
J	K	Q	S	R
0	0	0	0	X
0	1	0	0	X
1	0	0	1	0
1	1	0	1	0
0	1	1	0	1
1	1	1	0	1
0	0	1	X	0
1	0	1	X	0

4. For realisation of JK flip-flop from SR flip-flop, if $J=1, K=0$ & present state is 0 (i.e. $Q(n)=0$) then excitation input will be _____

- a) $S=0, R=1$
- b) $S=X, R=0$
- c) $S=1, R=0$
- d) $S=1, R=1$

Answer: c

Explanation: If $J=1, K=0$ & present state is 0 (i.e. $Q(n)=0$) then next state will be 1 which

results excitation inputs as $S=1$ & $R=0$.

FF data inputs		Output	S-R FF inputs	
J	K	Q	S	R
0	0	0	0	X
0	1	0	0	X
1	0	0	1	0
1	1	0	1	0
0	1	1	0	1
1	1	1	0	1
0	0	1	X	0
1	0	1	X	0

5. For realisation of SR flip-flop from JK flip-flop, the excitation input will be obtained from _____

- a) S and R
- b) R input
- c) J and K input
- d) D input

Answer: c

Explanation: It is the reverse process of SR flip-flop to JK flip-flop. So, for realisation of SR flip-flop from JK flip-flop, the excitation input will be obtained from J and K.

6. For realisation of SR flip-flop from JK flip-flop, if $S=1$, $R=0$ & present state is 0 then next state will be _____

- a) 1
- b) 0
- c) Don't care
- d) Toggle

Answer: a

Explanation: For JK flip-flop to SR flip-flop, if $S=1$, $R=0$ & present state is 0 then next state

will be 1 because next stage is complement of present stage.

<u>S-R Inputs</u>		<u>Outputs</u>		<u>J-K Inputs</u>	
S	R	Q _n	Q _{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	Invalid		Dont care	
1	1	Invalid		Dont care	

7. For realisation of SR flip-flop from JK flip-flop, if S=1, R=0 & present state is 0 then the excitation input will be _____

- a) J=1, K=1
- b) J=X, K=1
- c) J=1, K=X
- d) J=0, K=0

Answer: c

Explanation: For realisation of SR flip-flop from JK flip-flop, if S=1, R=0 & present state is 0 then the excitation input will be J=1, K=X.

<u>S-R Inputs</u>		<u>Outputs</u>		<u>J-K Inputs</u>	
S	R	Q _n	Q _{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	Invalid		Dont care	
1	1	Invalid		Dont care	

8. The K-map simplification for realisation of SR flip-flop from JK flip-flop is _____

- a) J=1, K=0
- b) J=R, K=S
- c) J=S, K=R
- d) J=0, K=1

Answer: c

Explanation: The K-map simplification for realisation of SR flip-flop from JK flip-flop is given by: $J=S$, $K=R$.

9. For realisation of D flip-flop from SR flip-flop, the external input is given through _____

- a) S
- b) R
- c) D
- d) Both S and R

Answer: c

Explanation: For realisation of D flip-flop from SR flip-flop, S and R are the actual inputs of the flip flop which is connected together via NOT gate and it is called external input as D.

10. For D flip-flop to JK flip-flop, the characteristics equation is given by _____

- a) $D=JQ(p)'+Q(p)K'$
- b) $D=JQ(p)'+KQ(p)'$
- c) $D=JQ(p)+Q(p)K'$
- d) $D=J'Q(p)+Q(p)K$

Answer: a

Explanation: A characteristic equation is needed when a specific gate requires a specific output in order to satisfy the truth table. For D flip-flop to JK flip-flop, the characteristics equation is given by $D=JQ(p)'+Q(p)K'$.
